

Gerard Baholli

14/11/1994

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PROFESSIONAL EXPERIENCE

- **Metaverso S.r.l.** Asolo (TV), IT
Software Developer Oct. 2023 - Present
 - Developed an interactive 3D visualization platform for scientific datasets (HDF5/FITS) with real-time rendering and VR support. Implemented the Unity client architecture, including project/file management, asynchronous data processing via REST APIs, JSON/MessagePack serialization, and caching of processed datasets. Created the visual-mapping features and a modular UI fully integrated with the rendering system, optimized for large datasets.
 - Developed custom 3D configurators using Three.js and TypeScript by analysing each product, defining configurable elements, and designing efficient mapping/texture strategies. Implemented performance optimizations (instanced meshes, async loading, memory-aware texture caching).
 - Developed interactive front-end experiences in Unreal Engine using Blueprint-based logic and UI systems. Built server-authoritative multi-user applications with reliable replication, synchronized state, and real-time communication.
- **Business Reply S.r.l.** Silea (TV), IT
Software Developer Apr. 2022 - Apr. 2023
 - Developed and optimized PL/SQL procedures for Oracle databases (on-premise and cloud), managing data processing workflows and improving query performance for large datasets.
 - Created data extraction and reporting pipelines, generating management reports through Oracle BI Publisher based on custom queries and business rules.
 - Built and maintained Oracle APEX applications for internal reporting and data visualization, enhancing usability and supporting business operations.
- **Aton S.p.A.** Villorba (TV), IT
Software Developer Dec. 2018 - Jun. 2019
 - Developed a warehouse stock monitoring system based on RFID technology, implementing the front-end in Angular and back-end logic in JavaScript.

EDUCATION

- **University of Milan** Milan, IT
Master's Degree in Computer Science Sept. 2019 - Oct. 2022
- **Ca' Foscari University of Venice** Venice, IT
Bachelor's Degree in Computer Science Sept. 2013 - Dec. 2018

PROJECTS

- **Master Thesis:** Human Activity Recognition using *OpenPose* and *TensorFlow* in Python.
- **Augmented Reality:** iOS AR application developed with *AR Foundation* and Unity (C#).
- **GPU Computing:** Parallel CUDA implementation of the Quickhull algorithm.
- **Game AI:** F1 driver AI using Behavior Trees and State Machines in Unity (C#).
- **Distributed Systems:** Token-based pollution detection system using *gRPC* in Java.

SKILLS

- **High-level:** C#, JavaScript/TypeScript, Java, Python.
- **Low-level:** C, CUDA.
- **Database:** SQL, PL/SQL, Oracle, PostgreSQL.
- **3D Engines & Frameworks:** Unity Engine, Three.js, Unreal Engine (Blueprints).
- **Tools & Versioning:** Git, Perforce.

LANGUAGES

Italian: Native

English: Level B2

Albanian: Native